

Blueprint for the Ternary Quartic Rank-4 Proof

2026-04-13

1 Statement and setup

We work in dehomogenized coordinates: a ternary quartic is viewed as a quartic polynomial in two affine variables x_0, x_1 . We use $\mathbb{R}[x_0, x_1]_{\leq k}$ to denote polynomials in these variables with degree at most k , and

$$\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\} = \text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\} \subset \mathbb{R}[x_0, x_1]_{\leq 2}$$

for the homogeneous quadratic subspace. The optimization variable is a rank-4 Burer–Monteiro factor

$$u = (u_0, u_1, u_2, u_3) \in \mathbb{R}[x_0, x_1]_{\leq 2}^4.$$

This is the same kind of vector-valued polynomial object written as $\mathbf{u}$ in the paper; here we suppress the boldface to keep the formulas short. All tangent directions are subject to the same quadratic degree bound.

Definition 1.1 (Admissibility). An *admissible point* or *admissible direction* is simply an element of $\mathbb{R}[x_0, x_1]_{\leq 2}^4$.

Definition 1.2 (Basic maps). For $u, v \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$ and $p \in \mathbb{R}[x_0, x_1]_{\leq 4}$, define

$$\sigma(u) = \sum_{i=0}^3 u_i^2, \quad A_u(v) = \sum_{i=0}^3 u_i v_i, \quad r(u, p) = \sigma(u) - p.$$

Fix a positive-definite bilinear form B on $\mathbb{R}[x_0, x_1]$. We say that u is a first-order critical point of p if

$$B(A_u(v), r(u, p)) = 0 \quad \text{for all } v \in \mathbb{R}[x_0, x_1]_{\leq 2}^4,$$

and a second-order critical point if, in addition,

$$B(\sigma(v), r(u, p)) + 2B(A_u(v), A_u(v)) \geq 0 \quad \text{for all } v \in \mathbb{R}[x_0, x_1]_{\leq 2}^4.$$

Definition 1.3 (SOS quartic target). An *SOS quartic* is a polynomial $p \in \mathbb{R}[x_0, x_1]_{\leq 4}$ admitting a finite representation

$$p = \sum_{j=1}^k q_j^2, \quad q_j \in \mathbb{R}[x_0, x_1]_{\leq 2}.$$

Definition 1.4 (Relation map and exact affine relations). For fixed $u \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$, define the relation map

$$\rho_u : \mathbb{R}^4 \rightarrow \mathbb{R}[x_0, x_1], \quad \rho_u(c) = \sum_{i=0}^3 c_i u_i.$$

Its kernel is the space of scalar linear dependences among the four coordinates of u .

Define the *exact affine relation space*

$$\mathcal{E}(u) = \{c \in \mathbb{R}^4 : \rho_u(c) \in \mathbb{R}[x_0, x_1]_{\leq 1}\}.$$

Equivalently, $\mathcal{E}(u)$ is the kernel of the map taking $\rho_u(c)$ to its homogeneous quadratic coefficients.

Definition 1.5 (Auxiliary notation). For $q \in \mathbb{R}[x_0, x_1]_{\leq 2}$, write

$$q = q_{\leq 1} + q_{\text{hom}} \quad \text{with } q_{\leq 1} \in \mathbb{R}[x_0, x_1]_{\leq 1}, \quad q_{\text{hom}} \in \text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$$

for its affine part and homogeneous quadratic part.

For $c \in \mathbb{R}^4$ and a polynomial h , write

$$c \odot h = (c_0h, \dots, c_3h)$$

whenever this vector belongs to $\mathbb{R}[x_0, x_1]_{\leq 2}^4$.

For $f \in \mathbb{R}[x_0, x_1]$, write

$$f = \sum_{a,b \geq 0} f_{ab} x_0^a x_1^b.$$

Thus f_{ab} denotes the coefficient of $x_0^a x_1^b$ in f .

If $x_0 \in \rho_u(\mathcal{E}(u))$, define $\mathcal{T}(u) \subseteq \text{span}\{1, x_1\}$ to be the set of affine tails $\alpha + \beta x_1$ for which there exists $c \in \mathcal{E}(u)$ such that $\rho_u(c)$ has x_0 -coefficient 0 and

$$\rho_u(c) - (\alpha + \beta x_1) \in \text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}.$$

Remark 1.6. The main theorem is that every admissible rank-4 SOCP over an SOS quartic has zero residual:

$$r(u, p) = 0.$$

Equivalently, there are no spurious second-order critical points in the rank-4 ternary quartic problem.

2 The certificate mechanism

The proof uses the same basic certificate as in the univariate case. For each square q^2 in an SOS decomposition, we write q^2 as a term in the image of a linearized map together with sums of squares of directions in $\text{Ker}(A_u)$. The SOCP inequalities make every kernel square pair nonnegatively with the residual, and this leaves only the zero residual.

Lemma 2.1 (Abstract image-plus-kernel-square certificate). *Let $u \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$ be a SOCP for an SOS quartic p . Assume that*

$$p = q_{\text{img}} + \sum_{m=1}^N \sigma(w^{(m)}),$$

where there is an admissible point u_{img} such that $q_{\text{img}} \in \text{Im}(A_{u_{\text{img}}})$, each $w^{(m)} \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$ satisfies $A_u(w^{(m)}) = 0$, and $r(u, p)$ is orthogonal to $\text{Im}(A_{u_{\text{img}}})$. Then $r(u, p) = 0$.

Proof. Set $r = r(u, p)$. The image term pairs to zero:

$$B(q_{\text{img}}, r) = 0.$$

Indeed, this is part of the hypothesis: q_{img} lies in $\text{Im}(A_{u_{\text{img}}})$, and r is orthogonal to that image. For each kernel direction $w = w^{(m)}$, the Hessian inequality gives

$$B(\sigma(w), r) + 2B(A_u(w), A_u(w)) \geq 0.$$

Since $A_u(w) = 0$, this reduces to $B(\sigma(w), r) \geq 0$. Summing over m yields

$$B(p, r) = B(q_{\text{img}}, r) + \sum_{m=1}^N B(\sigma(w^{(m)}), r) \geq 0.$$

On the other hand, first-order criticality with $v = u$ gives

$$B(\sigma(u), r) = 0.$$

Therefore

$$B(r, r) = B(\sigma(u) - p, r) = -B(p, r) \leq 0.$$

Positive-definiteness of B implies $B(r, r) \geq 0$, hence $B(r, r) = 0$, and therefore $r = 0$. $\square$

Corollary 2.2 (Image case). *If $p \in \text{Im}(A_u)$, then $r(u, p) = 0$.*

Proof. Apply Lemma 2.1 with $u_{\text{img}} = u$ and no kernel-square terms. The first-order condition says exactly that $r(u, p)$ is orthogonal to $\text{Im}(A_u)$. $\square$

Corollary 2.3 (Constant scalar relation). *If there exists $0 \neq c \in \mathbb{R}^4$ such that $\rho_u(c) = 0$, then $r(u, p) = 0$ for every SOS quartic p .*

Proof. Write $p = \sum_j q_j^2$ as in Definition 1.3. For each j , define

$$w^{(j)} = (c_0 q_j, \dots, c_3 q_j) \in \mathbb{R}[x_0, x_1]_{\leq 2}^4.$$

Then $A_u(w^{(j)}) = q_j \rho_u(c) = 0$, while

$$\sigma(w^{(j)}) = \left(\sum_{i=0}^3 c_i^2 \right) q_j^2.$$

Set $\lambda = \sum_i c_i^2 > 0$ and $\tilde{w}^{(j)} = \lambda^{-1/2} w^{(j)}$. Then $A_u(\tilde{w}^{(j)}) = 0$ and $\sigma(\tilde{w}^{(j)}) = q_j^2$. Therefore

$$p = \sum_j \sigma(\tilde{w}^{(j)}),$$

so Lemma 2.1 applies with no image term. $\square$

Proposition 2.4 (Equivariance under affine changes and orthogonal mixing). *The properties of being admissible, being an SOS quartic, being a SOCP, and having vanishing residual are preserved by*

1. *invertible affine changes of variables in x_0, x_1 , and*
2. *orthogonal changes of basis in the four-dimensional coefficient space of the factor coordinates.*

Proof. For an invertible affine change of variables e in x_0, x_1 , set $u'_i = e(u_i)$, $p' = e(p)$, and transport the bilinear form by

$$B'(f, g) = B(e^{-1}f, e^{-1}g).$$

Because e preserves polynomial degree, $u' \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$ whenever $u \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$. Moreover

$$\sigma(u') = e(\sigma(u)), \quad A_{u'}(v') = e(A_u(v)), \quad r(u', p') = e(r(u, p)).$$

Applying e^{-1} to the first- and second-order conditions shows that (u, p, B) is a SOCP exactly when (u', p', B') is. Since e is invertible, $r(u, p) = 0$ if and only if $r(u', p') = 0$.

For an orthogonal matrix $M \in O(4)$, let $u' = Mu$ and $v' = Mv$. Then orthogonality gives

$$\sigma(u') = \sigma(u), \quad A_{u'}(v') = A_u(v),$$

so the criticality inequalities are unchanged. The residual is literally the same polynomial:

$$r(u', p) = \sigma(u') - p = \sigma(u) - p = r(u, p).$$

Thus admissibility, the SOCP conditions, SOS-ness of p , and the equation $r(u, p) = 0$ are preserved under both operations. $\square$

3 Structural closure theorems

The full proof uses five structural results.

Theorem 3.1 (Span-three certificate). *Assume there exist $c^{(0)}, c^{(1)}, c^{(2)} \in \mathbb{R}^4$ such that*

$$\rho_u(c^{(0)}) = 1, \quad \rho_u(c^{(1)}) = x_0, \quad \rho_u(c^{(2)}) = x_1.$$

Then $r(u, p) = 0$ for every SOS quartic p .

Proof. Fix a quadratic $q \in \mathbb{R}[x_0, x_1]_{\leq 2}$. Write

$$q = q_{\leq 1} + q_{\text{hom}}, \quad q_{\text{hom}} = x_0 a + x_1 b$$

with $a, b \in \mathbb{R}[x_0, x_1]_{\leq 1}$. Then

$$A_u(c \odot h) = h \rho_u(c).$$

Define the raw kernel direction

$$w_{\text{raw}} = (-c^{(0)}) \odot q_{\text{hom}} + c^{(1)} \odot a + c^{(2)} \odot b.$$

By construction,

$$A_u(w_{\text{raw}}) = -q_{\text{hom}} + x_0 a + x_1 b = 0.$$

Now expand $\sigma(w_{\text{raw}})$. The only degree-4 contribution comes from the square of the first summand:

$$\sigma(w_{\text{raw}})_{\deg 4} = \left(\sum_{i=0}^3 (c_i^{(0)})^2 \right) q_{\text{hom}}^2.$$

All other squares or cross terms contain at least one factor a or b , hence have degree at most 3. Set

$$\lambda = \sum_i (c_i^{(0)})^2 > 0, \quad w = \lambda^{-1/2} w_{\text{raw}}.$$

Then $A_u(w) = 0$ and

$$\sigma(w) = q_{\text{hom}}^2 + h_3 \quad \text{for some } h_3 \in \mathbb{R}[x_0, x_1]_{\leq 3}.$$

Also,

$$q^2 - q_{\text{hom}}^2 = (q_{\leq 1} + q_{\text{hom}})^2 - q_{\text{hom}}^2 = q_{\leq 1}(2q_{\text{hom}} + q_{\leq 1}) \in \mathbb{R}[x_0, x_1]_{\leq 3},$$

so $q^2 - \sigma(w) \in \mathbb{R}[x_0, x_1]_{\leq 3}$.

Every cubic polynomial lies in $\text{Im}(A_u)$. Indeed, if $g \in \mathbb{R}[x_0, x_1]_{\leq 3}$, write

$$g = h_0 + x_0 h_1 + x_1 h_2 \quad \text{with } h_0 \in \mathbb{R}[x_0, x_1]_{\leq 2}, \quad h_1, h_2 \in \mathbb{R}[x_0, x_1]_{\leq 1}.$$

Then

$$g = A_u(c^{(0)} \odot h_0 + c^{(1)} \odot h_1 + c^{(2)} \odot h_2)$$

because $\rho_u(c^{(0)}) = 1$, $\rho_u(c^{(1)}) = x_0$, and $\rho_u(c^{(2)}) = x_1$. Therefore q^2 is a term in $\text{Im}(A_u)$ plus one kernel square.

Applying this construction to each quadratic summand in an SOS decomposition of p and summing gives the decomposition in Lemma 2.1. Hence $r(u, p) = 0$. $\square$

Theorem 3.2 (Mixed-affine theorem). *Assume there exist $c^{(0)}, c^{(1)} \in \mathbb{R}^4$ such that*

$$\rho_u(c^{(0)}) = 1, \quad \rho_u(c^{(1)}) = t + ax_0 + bx_1,$$

with $a^2 + b^2 \neq 0$. Then $r(u, p) = 0$ for every SOS quartic p .

Proof. By Proposition 2.4, we may apply an affine change of variables sending $t + ax_0 + bx_1$ to x_0 . Thus it is enough to treat the normalized case with exact relations for 1 and x_0 .

Choose two further relations whose homogeneous quadratic parts span the complementary two-dimensional subspace $P \subset \text{span}_{\mathbb{R}}\{x_0^2, x_0 x_1, x_1^2\}$. The plane normalization argument reduces P , after orthogonal mixing in the last two factor coordinates and a linear change in x_1 , to one of the following normal forms.

1. The surjective case: P is an invertible basis of either $\text{span}\{x_0^2, x_1^2\}$ or $\text{span}\{x_0 x_1, x_0^2 - x_1^2\}$. Then every quartic monomial lies in $\text{Im}(A_u)$: the relation 1 gives all monomials of degree ≤ 2 , the relation x_0 gives the missing cubic monomials with one x_0 factor, and the two quadratic relations give the remaining cubic and quartic monomials. Hence $\mathbb{R}[x_0, x_1]_{\leq 4} \subseteq \text{Im}(A_u)$, so Corollary 2.2 applies.

2. The rank-14 case: P is an invertible basis of $\text{span}\{x_0x_1, x_1^2\}$. Then the quotient $\mathbb{R}[x_0, x_1]_{\leq 4}/\text{Im}(A_u)$ is one-dimensional, generated by x_0^4 . An explicit kernel direction

$$w = (-c_2) \odot (x_0x_1) + c_3 \odot (x_0^2)$$

from the two plane relations c_2, c_3 . Its square is congruent, modulo $\text{Im}(A_u)$, to a nonzero multiple of x_0^4 . So the missing class is supplied by one kernel square.

3. The rank-13 case: P is an invertible basis of $\text{span}\{x_0^2, x_0x_1\}$. Then the only missing quotient classes are x_1^3 and x_1^4 . They are supplied by the kernel family

$$w_\ell = (-c_2) \odot (x_1\ell) + c_3 \odot (x_0\ell), \quad \ell = \mu + \nu x_1.$$

Varying $\ell \in \text{span}\{1, x_1\}$ gives two independent corrections modulo $\text{Im}(A_u)$, one for each missing quotient class.

In each normal form, every square q^2 is therefore a term in $\text{Im}(A_u)$ plus finitely many kernel squares. Summing over an SOS decomposition of p and applying Lemma 2.1 gives $r(u, p) = 0$. Finally, transport this back through the affine change of variables using Proposition 2.4. $\square$

Theorem 3.3 (Surjective homogeneous-basis theorem). *Assume there exists $c^{(0)} \in \mathbb{R}^4$ with $\rho_u(c^{(0)}) = 1$, and three further relations $d^{(0)}, d^{(1)}, d^{(2)} \in \mathbb{R}^4$ whose homogeneous quadratic parts form an invertible 3×3 matrix in the basis (x_0^2, x_0x_1, x_1^2) . Then $r(u, p) = 0$ for every SOS quartic p .*

Proof. Invertibility of the homogeneous coefficient matrix lets us solve back for exact relations with homogeneous parts exactly x_0^2, x_0x_1 , and x_1^2 . Since 1 is already an exact relation, every quartic monomial lies in $\text{Im}(A_u)$:

- degree ≤ 2 monomials come from multiplying the relation 1 by a quadratic,
- cubic monomials come from multiplying one of x_0^2, x_0x_1, x_1^2 by a linear polynomial,
- quartic monomials come from multiplying one of x_0^2, x_0x_1, x_1^2 by a quadratic monomial.

Thus every quartic p belongs to $\text{Im}(A_u)$, and Corollary 2.2 applies. $\square$

Theorem 3.4 (Low-affine plane theorem). *Assume there are exact relations for x_0 and x_1 , together with two more quadratic relations whose homogeneous parts span a two-dimensional subspace of $\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$. Then every SOS quartic p satisfies $r(u, p) = 0$.*

Proof. Fix exact relation vectors $c^{(0)}, c^{(1)} \in \mathbb{R}^4$ with $\rho_u(c^{(0)}) = x_0$ and $\rho_u(c^{(1)}) = x_1$. Let $d^{(2)}, d^{(3)} \in \mathbb{R}^4$ be the two additional quadratic relations. After subtracting suitable multiples of x_0 and x_1 , we may assume

$$\rho_u(d^{(2)}) = q_2, \quad \rho_u(d^{(3)}) = q_3, \quad q_\nu = a_\nu x_0^2 + b_\nu x_0x_1 + c_\nu x_1^2.$$

Set $A = b_2c_3 - c_2b_3$, $B = a_2c_3 - c_2a_3$, and $C = a_2b_3 - b_2a_3$. Since q_2 and q_3 are linearly independent, $(A, B, C) \neq (0, 0, 0)$. By Proposition 2.4, it is enough to classify the plane $P = \text{span}\{q_2, q_3\} \subseteq \text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$ after invertible linear changes of (x_0, x_1) .

Step 1: normal forms for P .

Assume first that $A = 0$, so $b_2c_3 - c_2b_3 = 0$. If $B \neq 0$, shear $(x_0, x_1) \mapsto (x_0, x_1 + tx_0)$ with $t = -C/(2B)$. The new x_0x_1 coefficient is $b'_\nu = b_\nu + 2tc_\nu$. When $c_\nu \neq 0$, the identity $A = 0$ gives $b_2/c_2 = b_3/c_3$, hence $C = (b_\nu/c_\nu)B$, so $b'_\nu = b_\nu - c_\nu C/B = 0$. When $c_\nu = 0$, $A = 0$ already forces $b_\nu = 0$, so again $b'_\nu = 0$. Thus both transformed quadratics have no x_0x_1 term, hence the transformed plane is $\text{span}\{x_0^2, x_1^2\}$.

If instead $B = 0$, then $C \neq 0$. We claim that $c_2 = c_3 = 0$. Indeed, if $c_2 \neq 0$, then $A = 0$ and $B = 0$ imply $b_3 = (c_3/c_2)b_2$ and $a_3 = (c_3/c_2)a_2$, so $C = a_2b_3 - b_2a_3 = 0$, a contradiction. The same argument rules out $c_3 \neq 0$. Therefore $q_2, q_3 \in \text{span}\{x_0^2, x_0x_1\}$, and $C \neq 0$ says that their coefficient matrix in the basis (x_0^2, x_0x_1) is invertible. So $P = \text{span}\{x_0^2, x_0x_1\}$, which becomes $\text{span}\{x_0x_1, x_1^2\}$ after exchanging x_0 and x_1 .

Now assume $A \neq 0$. Shearing by $t = -B/A$ gives new coefficients a'_ν, b'_ν, c'_ν satisfying

$$Aa'_\nu + \left(C - \frac{B^2}{A}\right)c'_\nu = 0 \quad (\nu = 2, 3).$$

Write $D = C - B^2/A$. If $D = 0$, then $a'_\nu = 0$ for both ν , so $P' = \text{span}\{x_0x_1, x_1^2\}$. If $D/A > 0$, scale $x_1 \mapsto sx_1$ with $s = \sqrt{D/A}$; then the relation above becomes $a''_\nu + c''_\nu = 0$, so every transformed quadratic has the form $b''_\nu x_0x_1 + a''_\nu(x_0^2 - x_1^2)$, and $P'' = \text{span}\{x_0x_1, x_0^2 - x_1^2\}$. If $D/A < 0$, scale $x_1 \mapsto sx_1$ with $s = \sqrt{(-D)/A}$; then $a''_\nu - c''_\nu = 0$, so every transformed quadratic has the form $b''_\nu x_0x_1 + a''_\nu(x_0^2 + x_1^2)$, and $P'' = \text{span}\{x_0x_1, x_0^2 + x_1^2\}$.

Thus, after an invertible linear change, P is one of

$$\text{span}\{x_0^2, x_1^2\}, \quad \text{span}\{x_0x_1, x_1^2\}, \quad \text{span}\{x_0x_1, x_0^2 + x_1^2\}, \quad \text{span}\{x_0x_1, x_0^2 - x_1^2\}.$$

Step 2: all nonconstant monomials lie in $\text{Im}(A_u)$.

In every normal form we still have exact relations for x_0 and x_1 , so $x_0 = A_u(c^{(0)} \odot 1)$ and $x_1 = A_u(c^{(1)} \odot 1)$. Hence $x_0^2 = A_u(c^{(0)} \odot x_0)$ and $x_0x_1 = A_u(c^{(0)} \odot x_1) = A_u(c^{(1)} \odot x_0)$. In the first two normal forms, x_1^2 is already an exact relation. In the last two normal forms, we have an exact relation for $x_0^2 \pm x_1^2$, and since $x_0^2 \in \text{Im}(A_u)$, we also get $x_1^2 \in \text{Im}(A_u)$. So every quadratic monomial lies in $\text{Im}(A_u)$. Multiplying the exact relations x_0, x_1 , and the exact quadratic relations by linear or quadratic polynomials shows that every monomial of degree 1, 2, 3, or 4 lies in $\text{Im}(A_u)$. The only possible missing quotient class is therefore the constant class.

Step 3: a kernel square produces the constant class.

Each normal form contains a nonzero exact quadratic divisible by x_1 : this is x_1^2 in the first two normal forms and x_0x_1 in the last two. Write that relation as $\rho_u(d) = x_1\ell$, where $\ell = x_1$ or $\ell = x_0$, and define

$$w_t = -c^{(1)} \odot (t\ell) + d \odot t.$$

Then $A_u(w_t) = -t\ell\rho_u(c^{(1)}) + t\rho_u(d) = 0$, so $w_t \in \text{Ker}(A_u)$. Its constant term comes only from the second summand, because every component of $-c^{(1)} \odot (t\ell)$ has positive degree. Thus $\sigma(w_t)_{00} = \alpha t^2$, where $\alpha = \sum_i d_i^2 > 0$.

If $p = \sum_j h_j^2$, then $p_{00} = \sum_j (h_j)_{00}^2 \geq 0$. Choosing $t = \sqrt{p_{00}/\alpha}$ makes $(p - \sigma(w_t))_{00} = 0$. Every other monomial of $p - \sigma(w_t)$ has positive degree, so Step 2 gives $p - \sigma(w_t) \in \text{Im}(A_u)$. Therefore $p = (p - \sigma(w_t)) + \sigma(w_t)$ is exactly an image term plus a kernel square, and Lemma 2.1 gives $r(u, p) = 0$. Finally, Proposition 2.4 transports the conclusion back to the original coordinates. $\square$

Theorem 3.5 (Affine-rank-one theorem after line normalization). *Assume $\dim \mathcal{E}(u) = 1$, that $\mathcal{E}(u)$ contains a nonconstant affine line but no constant relation, and normalize that line to x_0 . Let $\mathcal{T}(u)$ be the space of exact affine relations with zero x_0 coefficient, projected to their $(1, x_1)$ -tails. Then $\dim \mathcal{T}(u) \in \{0, 1, 2\}$, and in each of the three cases every SOS quartic satisfies $r(u, p) = 0$.*

Proof. After normalizing the unique exact affine line to x_0 , we can choose exact relations

$$r_{20} = \alpha_{20} + \beta_{20}x_1 + x_0^2, \quad r_{11} = \alpha_{11} + \beta_{11}x_1 + x_0x_1, \quad r_{02} = \alpha_{02} + \beta_{02}x_1 + x_1^2.$$

Let

$$T = \begin{pmatrix} \alpha_{20} & \beta_{20} \\ \alpha_{11} & \beta_{11} \\ \alpha_{02} & \beta_{02} \end{pmatrix}.$$

By definition,

$$\mathcal{T}(u) = \text{span}\{\alpha_{20} + \beta_{20}x_1, \alpha_{11} + \beta_{11}x_1, \alpha_{02} + \beta_{02}x_1\},$$

so $\dim \mathcal{T}(u) = \text{rank } T \in \{0, 1, 2\}$. We treat the three possibilities.

Case 1: $\dim \mathcal{T}(u) = 0$.

Here $T = 0$, so $r_{20} = x_0^2$, $r_{11} = x_0x_1$, and $r_{02} = x_1^2$. Hence we have exact relations for x_0, x_0^2, x_0x_1 , and x_1^2 . Every quartic monomial except 1 and x_1 therefore lies in $\text{Im}(A_u)$: any monomial with an x_0 -factor

is obtained by multiplying the exact relation x_0 or one of the exact quadratic relations by a low-degree polynomial, and the pure x_1 -monomials x_1^2, x_1^3, x_1^4 are handled the same way.

So only the constant and x_1 coefficients of p remain. Write $p = \sum_j h_j^2$, and set $s_0 = \sum_j (h_j)_{00}^2$ and $s_1 = \sum_j (h_j)_{00}(h_j)_{01}$, so $p_{00} = s_0$ and $p_{01} = 2s_1$. Choose $c_0, c_{11} \in \mathbb{R}^4$ with $\rho_u(c_0) = x_0$ and $\rho_u(c_{11}) = x_0x_1$, and for $\ell = c + dx_1$ define

$$w_\ell = -c_0 \odot (x_1\ell) + c_{11} \odot \ell.$$

Then $A_u(w_\ell) = -x_0(x_1\ell) + (x_0x_1)\ell = 0$, so $w_\ell \in \text{Ker}(A_u)$. If $\alpha = \sum_i c_{11,i}^2 > 0$ and $\beta = \sum_i c_{0,i}c_{11,i}$, then a direct coefficient check gives

$$\sigma(w_\ell)_{00} = \alpha c^2, \quad \sigma(w_\ell)_{01} = 2c(\alpha d - \beta c).$$

Choose $c = \sqrt{s_0/\alpha}$. If $c \neq 0$, let $d = s_1/(\alpha c) + \beta c/\alpha$; if $c = 0$, then $s_0 = 0$, hence $s_1 = 0$, so we take $d = 0$. With this choice, $\sigma(w_\ell)_{00} = p_{00}$ and $\sigma(w_\ell)_{01} = p_{01}$. Thus $(p - \sigma(w_\ell))_{00} = (p - \sigma(w_\ell))_{01} = 0$, and every remaining quartic monomial already lies in $\text{Im}(A_u)$. Hence $p - \sigma(w_\ell) \in \text{Im}(A_u)$, and Lemma 2.1 gives $r(u, p) = 0$.

Case 2: $\dim \mathcal{T}(u) = 1$.

Let $\tau = \rho + \eta x_1$ span $\mathcal{T}(u)$. Then there are scalars $\lambda_{20}, \lambda_{11}, \lambda_{02}$, not all zero, such that

$$r_{20} = \lambda_{20}\tau + x_0^2, \quad r_{11} = \lambda_{11}\tau + x_0x_1, \quad r_{02} = \lambda_{02}\tau + x_1^2.$$

Pure x_1 -tail. Suppose $\rho = 0$, and scale so that $\tau = x_1$. Then $r_{20} = \lambda_{20}x_1 + x_0^2$, $r_{11} = \lambda_{11}x_1 + x_0x_1$, and $r_{02} = \lambda_{02}x_1 + x_1^2$.

If $\lambda_{20} \neq 0$ or $\lambda_{11} \neq 0$, then $x_1 = \lambda_{20}^{-1}(r_{20} - x_0^2)$ or $x_1 = \lambda_{11}^{-1}(r_{11} - x_0x_1)$, so $x_1 \in \text{Im}(A_u)$. Hence every positive-degree monomial lies in $\text{Im}(A_u)$. To handle the constant class, we need one nonzero exact quadratic divisible by x_0 . If $\lambda_{20} = 0$, we can take x_0^2 ; if $\lambda_{11} = 0$, we can take x_0x_1 ; and if $\lambda_{20}\lambda_{11} \neq 0$, then

$$\lambda_{11}r_{20} - \lambda_{20}r_{11} = x_0(\lambda_{11}x_0 - \lambda_{20}x_1)$$

is such a relation. Writing $\rho_u(d) = x_0\ell$, the vector $w_t = -c_0 \odot (t\ell) + d \odot t$ satisfies $A_u(w_t) = 0$ and $\sigma(w_t)_{00} = \gamma t^2$ with $\gamma = \sum_i d_i^2 > 0$. Choosing $t = \sqrt{p_{00}/\gamma}$ removes the constant coefficient. Since every other monomial has positive degree, we get $p - \sigma(w_t) \in \text{Im}(A_u)$, and Lemma 2.1 applies.

If $\lambda_{20} = \lambda_{11} = 0$, then $\lambda_{02} \neq 0$, so $r_{20} = x_0^2$, $r_{11} = x_0x_1$, and $r_{02} = \lambda_{02}x_1 + x_1^2$. Every monomial with an x_0 -factor lies in $\text{Im}(A_u)$, and modulo $\text{Im}(A_u)$ we have $x_1^2 \equiv -\lambda_{02}x_1$, hence $x_1^3 \equiv \lambda_{02}^2x_1$ and $x_1^4 \equiv -\lambda_{02}^3x_1$. So every quartic is congruent modulo $\text{Im}(A_u)$ to a linear combination of 1 and x_1 . The same kernel family $w_\ell = -c_0 \odot (x_1\ell) + c_{11} \odot \ell$ from Case 1 still satisfies $\sigma(w_\ell)_{00} = \alpha c^2$ and $\sigma(w_\ell)_{01} = 2c(\alpha d - \beta c)$, so the same choice of c, d makes $(p - \sigma(w_\ell))_{00} = (p - \sigma(w_\ell))_{01} = 0$. Since only the 1 and x_1 classes can remain, this implies $p - \sigma(w_\ell) \in \text{Im}(A_u)$, and Lemma 2.1 again gives $r(u, p) = 0$.

Mixed constant-plus- x_1 tail. Suppose $\rho \neq 0$, and scale so that $\tau = 1 + bx_1$. Then

$$r_{20} = \lambda_{20}(1 + bx_1) + x_0^2, \quad r_{11} = \lambda_{11}(1 + bx_1) + x_0x_1, \quad r_{02} = \lambda_{02}(1 + bx_1) + x_1^2.$$

If $\lambda_{20} \neq 0$ or $\lambda_{11} \neq 0$, then $1 + bx_1 \in \text{Im}(A_u)$. Multiplying by x_1 gives $x_1 + bx_1^2 = (1 + bx_1)x_1 \in \text{Im}(A_u)$, while $x_1^2 = r_{02} - \lambda_{02}(1 + bx_1) \in \text{Im}(A_u)$. Subtracting bx_1^2 shows $x_1 \in \text{Im}(A_u)$, and then $1 = (1 + bx_1) - bx_1 \in \text{Im}(A_u)$. So every quartic monomial lies in $\text{Im}(A_u)$, hence $p \in \text{Im}(A_u)$, and Corollary 2.2 applies.

If $\lambda_{20} = \lambda_{11} = 0$, then $\lambda_{02} \neq 0$, so $r_{20} = x_0^2$, $r_{11} = x_0x_1$, and $r_{02} = \lambda_{02}(1 + bx_1) + x_1^2$. Again every monomial with an x_0 -factor lies in $\text{Im}(A_u)$, and now $x_1^2 \equiv -\lambda_{02}(1 + bx_1) \pmod{\text{Im}(A_u)}$. Reducing once more gives

$$x_1^3 \equiv \lambda_{02}^2b + (\lambda_{02}^2b^2 - \lambda_{02})x_1,$$

and reducing $x_1^4 = x_1^2 \cdot x_1^2$ the same way shows that x_1^4 is also congruent to a linear combination of 1 and x_1 . Hence every quartic is congruent modulo $\text{Im}(A_u)$ to a combination of 1 and x_1 , and the Case 1 kernel family again matches the coefficients p_{00} and p_{01} . Therefore $p - \sigma(w_\ell) \in \text{Im}(A_u)$, and Lemma 2.1 gives $r(u, p) = 0$.

Case 3: $\dim \mathcal{T}(u) = 2$.

Now the tail rows span $\text{span}\{1, x_1\}$. After reordering if necessary, we may write

$$r_{20} = a + bx_1 + x_0^2, \quad r_{11} = c + dx_1 + x_0x_1, \quad r_{02} = e + fx_1 + x_1^2,$$

with either $ad - bc \neq 0$ or $ad - bc = 0$ and one of the other 2×2 minors nonzero.

If $ad - bc \neq 0$, then $x_0^2, x_0x_1 \in \text{Im}(A_u)$, so $a + bx_1 = r_{20} - x_0^2 \in \text{Im}(A_u)$ and $c + dx_1 = r_{11} - x_0x_1 \in \text{Im}(A_u)$. The linear combinations

$$d(a + bx_1) - b(c + dx_1) = (ad - bc) \cdot 1,$$

$$c(a + bx_1) - a(c + dx_1) = (bc - ad) \cdot x_1$$

show that $1, x_1 \in \text{Im}(A_u)$, and then $x_1^2 = r_{02} - e \cdot 1 - fx_1 \in \text{Im}(A_u)$. So every quadratic monomial lies in $\text{Im}(A_u)$, hence every quartic monomial does too, and Corollary 2.2 gives $r(u, p) = 0$.

It remains to treat $ad - bc = 0$. First suppose $(a, b) \neq (0, 0)$. Then $(c, d) = g(a, b)$ for some $g \in \mathbb{R}$. Shearing by $(x_0, x_1) \mapsto (x_0, x_1 + gx_0)$ and replacing r_{11} by $r_{11} - gr_{20}$ gives

$$r'_{20} = a + bx_1 + x_0^2, \quad r'_{11} = x_0x_1, \quad r'_{02} = e' + f'x_1 + x_1^2,$$

where $e' = e - g^2a$ and $f' = f - g^2b$. Set $\Delta = a^2 - abf' + b^2e' = a^2 - abf + b^2e$.

If $\Delta \neq 0$, then $a + bx_1 = r'_{20} - x_0^2 \in \text{Im}(A_u)$, and $ax_1 + bx_1^2 = r'_{20}x_1 - x_0^2x_1 \in \text{Im}(A_u)$. Also

$$(be' - af') \cdot 1 + bx_1^2 = br'_{02} - f'r'_{20} + f'x_0^2 \in \text{Im}(A_u),$$

so subtracting gives $(af' - be') \cdot 1 + ax_1 \in \text{Im}(A_u)$. Combining this with $a + bx_1$ yields $\Delta \cdot 1 \in \text{Im}(A_u)$ and $\Delta \cdot x_1 \in \text{Im}(A_u)$. Since $\Delta \neq 0$, we conclude $1, x_1 \in \text{Im}(A_u)$, so all quartics lie in $\text{Im}(A_u)$.

If $\Delta = 0$ and $a \neq 0$, then $b \neq 0$ because otherwise $\Delta = a^2 \neq 0$. Translate $x_1 \mapsto y_1 + a/b$, so $y_1 = x_1 - a/b$. Then $r'_{20} = by_1 + x_0^2$, hence after scaling we obtain a relation $y_1 + \tilde{a}x_0^2$. Also

$$r'_{02} = \left(e' - \frac{a}{b}f' + \frac{a^2}{b^2}\right) + \left(f' - \frac{2a}{b}\right)y_1 + y_1^2.$$

The constant term is $\Delta/b^2 = 0$, so the translated third relation is either y_1^2 or $y_1 + \tilde{d}y_1^2$ after rescaling. This is one of the pure- x_1 endpoint families already handled in Case 2. If $a = 0$, then $\Delta = b^2e' = 0$ forces $e' = 0$, so $r'_{20} = bx_1 + x_0^2$, $r'_{11} = x_0x_1$, and $r'_{02} = f'x_1 + x_1^2$, which is again a Case 2 endpoint family.

Finally, suppose $(a, b) = (0, 0)$. Then $r_{20} = x_0^2$ is exact, and the tail rank condition forces the tails of r_{11} and r_{02} to be independent. Write $r_{11} = c + dx_1 + x_0x_1$ and $r_{02} = e + fx_1 + x_1^2$, and set $\Delta' = c^2 - cdf + d^2e$.

If $\Delta' \neq 0$, then $c + dx_1 = r_{11} - x_0x_1 \in \text{Im}(A_u)$ and $cx_1 + dx_1^2 = r_{11}x_1 - x_0x_1^2 \in \text{Im}(A_u)$. Also

$$(de - cf) \cdot 1 + dx_1^2 = dr_{02} - fr_{11} + fx_0x_1 \in \text{Im}(A_u),$$

so $(cf - de) \cdot 1 + cx_1 \in \text{Im}(A_u)$. Combining these gives $\Delta' \cdot 1 \in \text{Im}(A_u)$ and $\Delta' \cdot x_1 \in \text{Im}(A_u)$, hence $1, x_1 \in \text{Im}(A_u)$. So again every quartic lies in $\text{Im}(A_u)$.

If $\Delta' = 0$ and $d \neq 0$, translate by $y_1 = x_1 - c/d$. Then $r_{11} = dy_1 + x_0y_1 + (c/d)x_0$, so subtracting $(c/d)x_0$ and rescaling gives a relation $y_1 + \tilde{c}x_0y_1$. For r_{02} , the constant term after translation is

$$e - \frac{c}{d}f + \frac{c^2}{d^2} = \frac{\Delta'}{d^2} = 0,$$

so the translated third relation is either y_1^2 or $y_1 + \tilde{d}y_1^2$. This is again one of the pure- x_1 endpoint families from Case 2. If $d = 0$, then $\Delta' = c^2 = 0$, so $c = 0$, hence $r_{11} = x_0x_1$, and we are already in those endpoint families.

This exhausts the three possibilities for $\dim \mathcal{T}(u)$, so every SOS quartic p satisfies $r(u, p) = 0$. $\square$

4 The exact-affine classifier

We now explain the classifier that handles the linearly independent case.

Lemma 4.1 (Dimension bounds for exact affine relations). *Assume $\text{Ker}(\rho_u) = \{0\}$. Then*

$$1 \leq \dim \mathcal{E}(u) \leq 3.$$

Hence $\dim \mathcal{E}(u) \in \{1, 2, 3\}$.

Proof. By Definition 1.4, $\mathcal{E}(u)$ is the kernel of the homogeneous coefficient map

$$\mathbb{R}^4 \longrightarrow \text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\} \simeq \mathbb{R}^3.$$

Since the domain has dimension 4 and the codomain has dimension 3, rank-nullity gives $\dim \mathcal{E}(u) \geq 1$. On the other hand, because $\text{Ker}(\rho_u) = \{0\}$, the affine coefficient map on $\mathcal{E}(u)$ is injective, so $\dim \mathcal{E}(u) \leq \dim \mathbb{R}[x_0, x_1]_{\leq 1} = 3$. $\square$

Theorem 4.2 (Exact-affine dimension 3). *If $\text{Ker}(\rho_u) = \{0\}$ and $\dim \mathcal{E}(u) = 3$, then $r(u, p) = 0$.*

Proof. The affine coefficient map $\mathcal{E}(u) \rightarrow \mathbb{R}[x_0, x_1]_{\leq 1}$ is injective and both domain and codomain have dimension 3, so it is an isomorphism. Therefore there are exact relations for $1, x_0, x_1$. Apply Theorem 3.1. $\square$

Theorem 4.3 (Exact-affine dimension 2 with a constant relation). *If $\text{Ker}(\rho_u) = \{0\}$, $\dim \mathcal{E}(u) = 2$, and $1 \in \rho_u(\mathcal{E}(u))$, then $r(u, p) = 0$.*

Proof. Fix an exact constant relation 1. Since $\mathcal{E}(u)$ has dimension 2, there is a second independent exact affine relation $t + ax_0 + bx_1$ with $a^2 + b^2 \neq 0$. Apply Theorem 3.2. $\square$

Theorem 4.4 (Exact-affine dimension 2 without a constant relation). *If $\text{Ker}(\rho_u) = \{0\}$, $\dim \mathcal{E}(u) = 2$, and $1 \notin \rho_u(\mathcal{E}(u))$, then $r(u, p) = 0$.*

Proof. First extract an affine pair $x_0 + r_0, x_1 + r_1$ from $\mathcal{E}(u)$, then translate by $(x_0, x_1) \mapsto (x_0 - r_0, x_1 - r_1)$. After this translation, we have exact relations for x_0 and x_1 , and every remaining exact affine relation has zero linear part.

Consider the translated zero-linear-tail kernel. There are two possibilities.

1. Every such relation has zero constant term. Then the complementary quadratic relations are purely homogeneous. Their homogeneous parts span a two-dimensional subspace of $\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$, so Theorem 3.4 applies.
2. Some translated kernel relation has nonzero constant term. Then choose a pair of translated relations with constants 1 and 0. If their homogeneous parts are dependent, a linear combination produces a constant relation. Together with the exact relations for x_0 and x_1 , this gives exact relations for $1, x_0, x_1$, so Theorem 3.1 applies. If their homogeneous parts are independent, then we are in the constant-perturbed low-affine situation. The corresponding arguments analyze the same common-factor, cross, and diagonal normal forms as in Theorem 3.4, now with one relation shifted by 1, and again show that every square q^2 is an image term plus kernel squares.

Transporting back through the translation preserves residual vanishing by Proposition 2.4, so $r(u, p) = 0$. $\square$

Theorem 4.5 (Exact-affine dimension 1 with a constant relation). *If $\dim \mathcal{E}(u) = 1$ and $1 \in \rho_u(\mathcal{E}(u))$, then $r(u, p) = 0$.*

Proof. Let $c^{(0)}$ generate $\mathcal{E}(u)$ with $\rho_u(c^{(0)}) = 1$. Consider the subspace

$$K = \{c \in \mathbb{R}^4 : \text{the constant term of } \rho_u(c) \text{ is } 0\}.$$

Because $\rho_u(c^{(0)}) = 1$, the constant coefficient map is surjective, so $\dim K = 3$. Now map K to $\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$ by taking the homogeneous quadratic part of $\rho_u(c)$. This map is injective: if $c \in K$ maps to 0, then $\rho_u(c)$ is an exact affine relation with zero constant term. Since $\dim \mathcal{E}(u) = 1$ and $\mathcal{E}(u)$ is generated by the relation 1, the only such relation is 0. Therefore the image of K in $\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$ also has dimension 3, hence equals all of $\text{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$.

So we may choose $d^{(0)}, d^{(1)}, d^{(2)} \in K$ whose homogeneous parts are exactly x_0^2, x_0x_1 , and x_1^2 . Their lower-degree parts involve only x_0 and x_1 , so they are relations of the form

$$(\text{affine tail}) + x_0^2, \quad (\text{affine tail}) + x_0x_1, \quad (\text{affine tail}) + x_1^2.$$

The homogeneous coefficient matrix is the identity, so Theorem 3.3 applies and gives $r(u, p) = 0$. $\square$

Theorem 4.6 (Exact-affine dimension 1 without a constant relation). *If $\text{Ker}(\rho_u) = \{0\}$, $\dim \mathcal{E}(u) = 1$, and $1 \notin \rho_u(\mathcal{E}(u))$, then $r(u, p) = 0$.*

Proof. Extract the unique exact affine line and normalize it by an affine equivalence to x_0 . After this normalization, the remaining exact affine relations with zero x_0 -coefficient define the tail space $\mathcal{T}(u) \subseteq \text{span}\{1, x_1\}$. Its dimension is 0, 1, or 2. The three cases are exactly the three cases covered by Theorem 3.5, so $r(u, p) = 0$. $\square$

Theorem 4.7 (Exact-affine classifier). *If $\text{Ker}(\rho_u) = \{0\}$, then $r(u, p) = 0$ for every SOS quartic p and every SOCP u .*

Proof. By Lemma 4.1, $\dim \mathcal{E}(u) \in \{1, 2, 3\}$.

- If $\dim \mathcal{E}(u) = 3$, apply Theorem 4.2.
- If $\dim \mathcal{E}(u) = 2$, split according to whether $1 \in \rho_u(\mathcal{E}(u))$ and apply Theorems 4.3 and 4.4.
- If $\dim \mathcal{E}(u) = 1$, split according to whether $1 \in \rho_u(\mathcal{E}(u))$ and apply Theorems 4.5 and 4.6.

These cases exhaust the possibilities. $\square$

5 Proof of the main theorem

Theorem 5.1 (No spurious rank-4 SOCPs for ternary quartics). *Let p be an SOS quartic, let $u \in \mathbb{R}[x_0, x_1]_{\leq 2}^4$, and let B be a positive-definite bilinear form on $\mathbb{R}[x_0, x_1]$. If u is a second-order critical point of p , then*

$$r(u, p) = 0.$$

Proof. Split into two cases according to $\text{Ker}(\rho_u)$.

Case 1: $\text{Ker}(\rho_u) \neq \{0\}$. Then there exists $0 \neq c \in \mathbb{R}^4$ with $\rho_u(c) = 0$. By Corollary 2.3, $r(u, p) = 0$.

Case 2: $\text{Ker}(\rho_u) = \{0\}$. Then Theorem 4.7 applies directly and gives $r(u, p) = 0$.

This proves the theorem. $\square$

Remark 5.2. The proof has the same overall shape as the univariate argument but with a different algebraic middle layer. The univariate proof closes by a single divisibility argument on the common factor g . In the ternary quartic case, the multivariate replacement is the exact-affine classifier:

$$\text{dependent relation or } \dim \mathcal{E}(u) \in \{1, 2, 3\},$$

followed by the span-three, mixed-affine, low-affine, and affine-rank-one theorems described above.

6 Formalization Cross-Reference

table 1 collects the corresponding Lean declarations and source files.

Table 1: Formalization cross-reference for the statements used in this blueprint.

Blueprint item	Lean declaration(s) or source file(s)
Definition 1.2	<code>sigma</code> , <code>A</code> , <code>residual</code> , <code>IsSOCP</code>
Definition 1.3	<code>IsSOSQuartic</code>
Definition 1.4	<code>relationPoly</code> , <code>relationPolyLin</code> , <code>exactAffineSubmodule</code>
Lemma 2.1	<code>admissible_image_plus_cone_residual_eq_zero</code>
Corollary 2.3	<code>residual_eq_zero_of_constant_relation</code>
Proposition 2.4	<code>AffineSocpTransform.lean</code> , <code>FactorTransform.lean</code>
Theorem 3.1	<code>residual_eq_zero_of_contains_aff1</code>

Blueprint item	Lean declaration(s) or source file(s)
Theorem 3.2	<code>residual_eq_zero_of_relations_const_affineLine</code>
Theorem 3.3	<code>exact_relations_of_homQuadBasis_det,</code> <code>residual_eq_zero_of_relations_const_homQuadBasis_det</code>
Theorem 3.4	<code>RepresentativeLowAffine.lean</code>
Theorem 3.5	<code>residual_eq_zero_of_exactAffineDimOne_affineLine_tailRangeZero,</code> <code>residual_eq_zero_of_exactAffineDimOne_affineLine_tailRangeOne,</code> <code>residual_eq_zero_of_exactAffineDimOne_affineLine_tailRangeTwo</code>
Theorem 4.2	<code>residual_eq_zero_of_exactAffineDimThree</code>
Theorem 4.3	<code>residual_eq_zero_of_exactAffineDimTwo_contains_one</code>
Theorem 4.4	<code>residual_eq_zero_of_exactAffineDimTwo_noConst</code>
Theorem 4.5	<code>residual_eq_zero_of_exactAffineDimOne_contains_one</code>
Theorem 4.6	<code>residual_eq_zero_of_exactAffineDimOne_noConst</code>
Theorem 4.7	<code>residual_eq_zero_of_relationPolyLin_ker_eq_bot</code>
Theorem 5.1	<code>ternaryQuartic_rankFour_no_spurious_socp</code>